Akoizo Original Practice - 2027 Rotation Form 2

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Hydrophobic attraction is a new covalent bond?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Always

B. Only between leucines

C. Only in water

D. No

Answer: _____

2. (2 points) HA contributes to…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Attachment and fusion

B. Only genome replication

C. Only salt sensing

D. Only ATP storage

Answer: _____

3. (2 points) A strong annotation explains…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Only material cost

B. Identity, role, and evidence

C. Only a color

D. Only a teacher name

Answer: _____

4. (2 points) A disulfide links…

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Any two glycines

B. DNA bases

C. Two phosphate groups

D. Cysteines

Answer: _____

5. (2 points) Reassortment exchanges…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Genome segments

B. Only individual atoms

C. Only amino-acid colors

D. Only antibodies

Answer: _____

6. (2 points) A missing segment should be…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Ignored in numbering

B. Explicitly marked or qualified

C. Silently invented

D. Called observed

Answer: _____

7. (2 points) More geometric contacts prove greater stability?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Always

B. Only in cartoons

C. Only for enzymes

D. No

Answer: _____

8. (2 points) The 2009 outbreak is used here as…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. A historical case study

B. A current case count

C. A permanent event assignment

D. A diagnosis guide

Answer: _____

9. (2 points) Model review should use…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. An unrelated year automatically

B. The verified applicable rubric

C. Invented point values

D. Only appearance

Answer: _____

10. (4 points) Why does pH matter?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

Answer: _____
_____
_____

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

Answer: _____
_____
_____

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

Answer: _____
_____
_____

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

Answer: _____
_____
_____

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

Answer: _____
_____
_____

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

Answer: _____
_____
_____

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

Answer: _____
_____
_____

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

Answer: _____
_____
_____

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

Answer: _____
_____
_____

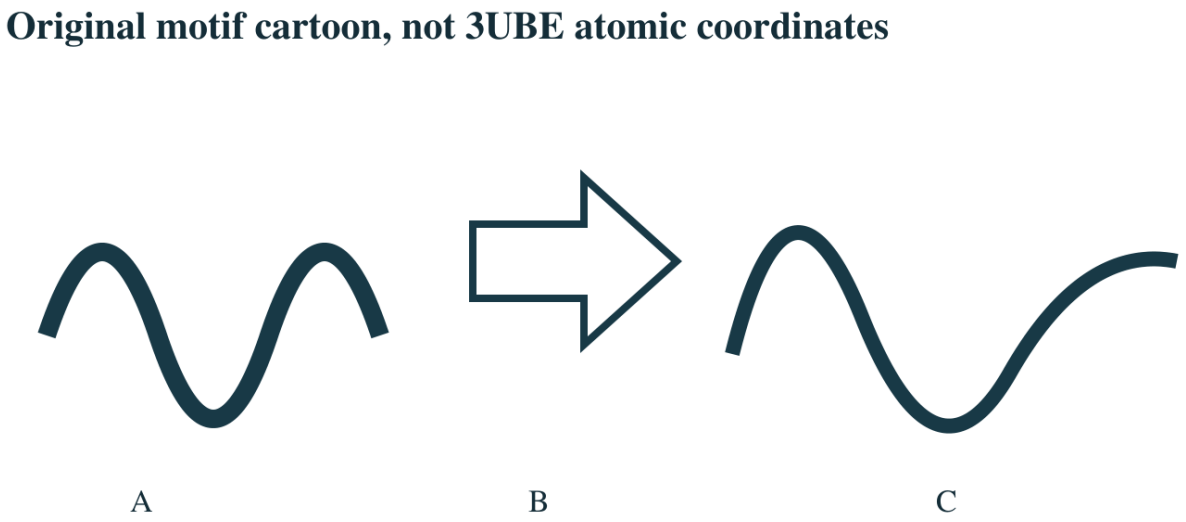
19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

Answer: _____
_____
_____

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Identify the secondary-structure motif convention represented by B.



F1. Backbone motifs and selected residue interval (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. How many residue positions are included in the selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

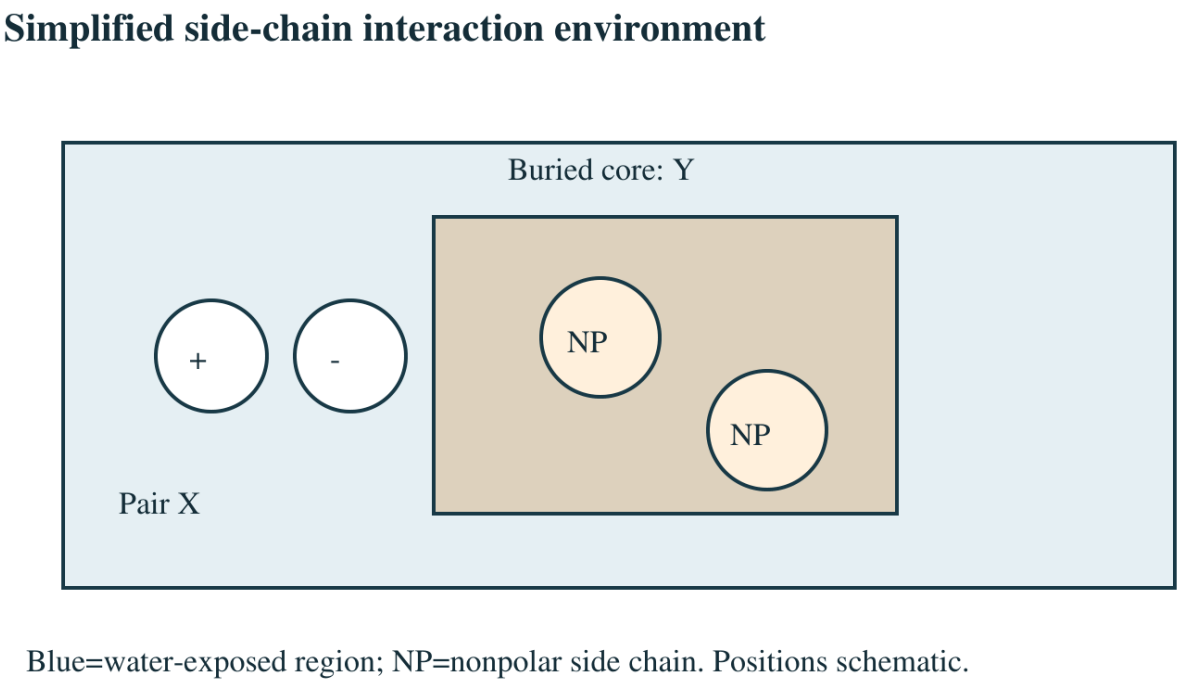
Answer: _____
_____
_____

23. (2 points) Use Figure F1. At2 cm per residue, what model backbone length represents that selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. What stabilizing interaction could occur at pair X? State one condition that affects it.



F2. Charge and burial in a folded protein (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Explain why nonpolar side chains in Y can be favorable in an aqueous environment.

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

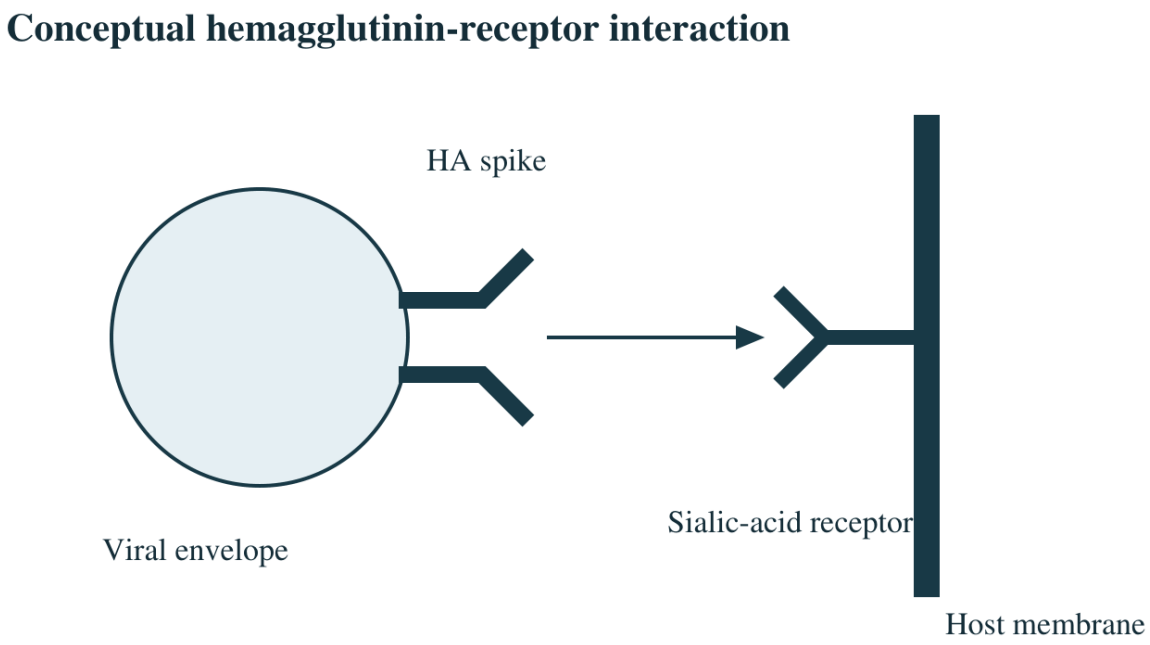
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Why could changing pH alter the displayed interaction at X?

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. [State/National] Which labeled viral feature directly participates in the depicted receptor attachment?



F3. Influenza attachment mechanism (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. [State/National] Predict how blocking the displayed receptor-binding interaction could affect this entry step.

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

29. (2 points) Use Figure F3. [State/National] Why does this attachment drawing not prove that a bound particle has completed membrane fusion?

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which recorded field distinguishes the intended chain from other chains in the same structure?

Illustrative JUDE selection record; not a captured application screen

Field	Recorded selection
Structure	3UBE
Chain	A
Residue range	153-252 inclusive
Display	Backbone cartoon
View action	Rotate camera
Coordinate check	Check missing residues before building

F4. Molecular-view selection record (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Does the recorded camera rotation change primary sequence or bonds? Explain.

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Why is the final coordinate check important even when the numbered interval is correctly recorded?

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____